

Comprehensive Delta Report: V00 vs V2

1. Summary of Changes

- Title change: V00 focuses on 'Neural Network Emulation', V2 reframes as 'Hybrid Neural Force Correction'.
- Core shift: From pure NN emulation → hybrid physics + NN correction system.
- Positioning improved: V2 is more grounded, realistic, and reviewer-friendly.
- Results interpretation refined: clearer framing of chaos and divergence.

2. Detailed Differences

Methodology: V2 clarifies hybrid design, adds gating, emphasizes NN used only in ~1% interactions.

Integrator description: V00 incorrectly labels IAS15 as symplectic; V2 corrects this and improves explanation.

Framing: V00 suggests NN could replace integrators; V2 explicitly avoids this claim.

Evaluation: V2 introduces 'divergence-rate proxy' clarification (more rigorous).

3. Results Comparison

- Numerical results (speedup ~2.09x, $\lambda \approx 0.156$) are IDENTICAL.
- Interpretation improved in V2 (more cautious, less overclaiming).
- Chaos explanation refined (proxy vs true Lyapunov).

4. Model Runs Consistency

- SAME experiments, SAME outputs.
- Only interpretation + wording changed.
- No evidence of new training or reruns.

5. Conclusions Evolution

V00: Slightly overclaims NN capability.

V2: Positions NN as assistive tool, not replacement.

- Stronger scientific credibility.
- Better alignment with reviewer expectations.

Final Verdict

V2 is a refinement, not a new experiment. It significantly improves scientific rigor, clarity, and acceptance probability without changing underlying results.